



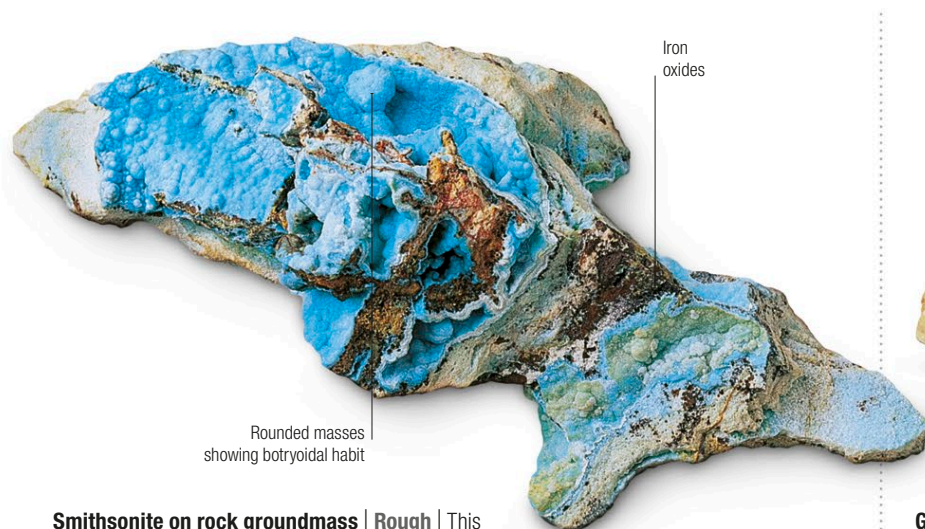
Smithsonite

△ Rectangular cabochon of smithsonite

Smithsonite can be various colours including yellow and pink, but blue-green is the most prized of all. Crystals are found occasionally – spectacular examples come from Tsumeb, Namibia – and sometimes faceted for collectors. Most gemstone material is cut *en cabochon* or carved into ornaments, but it is too soft for general wear as jewellery. Aside from its use as a gem, it is mined as a major source of zinc; it is thought that it may have provided the zinc component of brass in ancient metallurgy. One of the main sources is the Kelly Mine, New Mexico, USA.

Specification

Chemical name Zinc carbonate | **Formula** ZnCO_3 | **Colours** White, blue, green, yellow, brown, pink, lilac, colourless | **Structure** Hexagonal | **Hardness** 4–4.5 | **SG** 4.3–4.5 | **RI** 1.62–1.85 | **Lustre** Vitreous to pearly | **Streak** White | **Locations** Namibia, Zambia, Australia, Mexico, Germany, Italy, USA



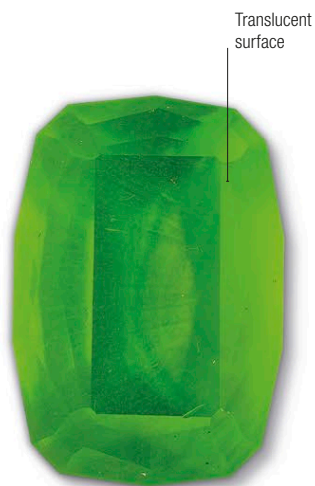
Smithsonite on rock groundmass | Rough | This smithsonite specimen exhibits layers of intense colour set in a groundmass of iron oxides. Its botryoidal habit (like bunches of grapes) is visible.



Greek specimen | Rough | This smithsonite from Avron, Attica, Greece, shows one a yellow hue rather than the more commonly seen blue green shades.



Cabochon | Cut | Smithsonite is brittle, soft, and easily abraded or chipped, as at the base of this example, but this cabochon is desirable for its intense blue colour.



Kelly specimen | Cut | This rare, faceted smithsonite, is an example of the superb-quality material that came from the Kelly Mine, New Mexico, USA.



Oval cabochon | Cut | The solid translucent material of this oval cabochon brings out the blue-green hue, the colour traditionally associated with smithsonite.

Changing names

Calamine to Smithson

Smithsonite was originally called by the umbrella name calamine – as in the anti-itch lotion used to treat skin problems, which contains the powdered mineral. The English chemist and mineralogist John Smithson discovered that calamine was in fact three different minerals, and smithsonite was named in his honour in 1832. The other two minerals were called hemimorphite and hydrozincite. Smithson also made the bequest that led to the establishment of the Smithsonian Institution.



James Smithson (1765–1829) Smithsonite is named after Smithson, the mineralogist who discovered it.